SERIES

Number Series + Letter & Alphanumeric Series

OAVS TGT | Reasoning (Verbal) | Topic 1 of the syllabus list

Syllabus line (Reasoning, 30 Marks): “The Verbal Reasoning topics like **Series**, Ranks, Direction, Arrangement, Coding, Decoding, Analogy and Classification/Odd Pair, Syllogism and Statement Conclusion.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2) reasoning sections. Every one of the 16 series questions found in these papers is solved here.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	How many questions, which patterns, priority order, index of all 16 PYQs
2	Foundation	Attack order, speed hints, reference tables (squares, cubes, primes, powers)
3	Number Series: 9 Types	Rule, how to spot, shortcut and step-by-step PYQ solutions
4	Letter & Alphanumeric Series	Position table, column method, wrong cluster, missing letters
5	Shortcuts and Traps	Speed methods and the mistakes that cost marks
6	Practice Set	28 original questions in exam pattern with answer key
7	Quick Revision	One-page cheat sheet for the last day

SCOPE: Figure/diagram series (shape sequences) belong to the non-verbal topics (Rule Detection, Completion of Pattern) and letter-cluster *analogy* belongs to Analogy/Coding. They are intentionally not repeated here.

CHAPTER 1

PYQ Analysis

What the five papers tell us about Series

Series questions found

Paper	Q. numbers	Count	Mix
English 2019	33, 37, 40	3	3 number series
English 2022	41, 56	2	2 letter / alphanumeric (find next)
Mathematics 2024	78, 80, 91, 99	4	2 number + 2 incorrect letter-cluster
Science 2024	73, 83	2	1 number + 1 odd letter-cluster
Social Studies 2023	3, 11, 14, 15, 20	5	3 number (incl. 1 wrong term) + 2 letter/alphanumeric
Total		**16**	**9 number + 7 letter/alphanumeric = about 3 per 30-question paper**

Pattern frequency and priority

Pattern	PYQs	Priority
Number: differences in AP (2nd difference constant)	5	★★★ Must
Letter / alphanumeric cluster: each position shifts by its own rule	6	★★★ Must
Number: differences in GP (diffs double)	1	★★ High
Number: 3rd-level pattern (2nd differences are squares)	1	★★ High
Number: alternating × and + with rising numbers	1	★★ High
Number: wrong term (×2 − n type)	1	★★ High
Repeating letter block with blanks	1	★★ High
Not yet asked but likely: ×n ± k, squares/cubes families, interleaved, Fibonacci-type, primes	0	★ Expected

Question formats

Format	PYQs	Example wording
Find the next term / ?	11	“What would be the next term in the following series”
Fill the blanks sequentially	1	“Select the combination of letters that when sequentially placed in the blanks…”
Wrong / odd term	4	“only one letter-cluster is incorrect” / “DOES NOT belong to the series” / “does NOT belong” (number)

PYQ INSIGHT: In **5 of the 9** number series the first difference alone is not enough; you must take the **difference of the differences**. The differences 5, 10, 15, 20 appeared in **both 2019 (Q37) and 2024 (Q78)** with different starting numbers. Revise this pattern first.

PYQ INSIGHT: All number series have 5 or 6 given terms and one blank at the end, with 4 options. Letter series are 5 to 6 clusters of 3 letters (or letter-number-letter). Target time: **40 seconds** per question.

Index of all 16 PYQs (solutions are in the chapters shown)

Paper / Q	Series	Rule	Ans	See
Eng 2019 Q37	9, 14, 24, 39, 59, ?	diffs +5 each time	84	3.2
Eng 2019 Q40	3, 17, 45, 87, 143, ?	diffs 14×1, 14×2…	213	3.2

Paper / Q	Series	Rule	Ans	See
Eng 2019 Q33	6, 18, 22, 110, 116, 812, ?	×3, +4, ×5, +6, ×7, +8	820	3.4
Maths 2024 Q78	11, 16, 26, 41, 61, ?	diffs 5, 10, 15, 20, 25	86	3.2
Maths 2024 Q91	16, 24, 40, 72, 136, ?	diffs double	264	3.2
Science 2024 Q83	112, 120, 130, 142, 156, ?	diffs +2 each	172	3.2
SST 2023 Q20	24, 39, 64, 99, 144, ?	diffs +10 each	199	3.2
SST 2023 Q3	33, 44, 59, 83, 123, ?	2nd diffs 2^2, 3^2, 4^2…	188	3.2
SST 2023 Q11	15, 27, 50, 95, 184, 366	×2 − 3, −4, −5…	366 wrong	3.9
Eng 2022 Q41	HSD, KPI, NMN, QJS, ?	+3, −3, +5	TGX	4.3
Eng 2022 Q56	H22N, I19J, K18G, N19E, ?	+1,+2,+3 | −3,−1,+1 | −4,−3,−2	R22D	4.4
SST 2023 Q15	FWCU-5, IZZR-15, LCWO-35, ?, RIQI-155	+3,+3,−3,−3 | ×2+5	OFTL-75	4.4
Maths 2024 Q80	XQS VOQ TMO RKM PIL NGI	each position −2	PIL	4.5
Maths 2024 Q99	SUX QSW OQT MOR KMP IKN	−2, −2, then −2 from T	QSW	4.5
Science 2024 Q73	AWL XQG SKD OEZ KYV GSR	−4, −6, −4	XQG	4.5
SST 2023 Q14	P_MR_E_FM_W_PF_R_E	block PFMRWE repeats	FWPREMW	4.6

CHAPTER 2

Foundation

Essential basics before the patterns

2.1 The attack order (use every time)

Step	Test	Time	If it works
1	Look at the growth: slow, doubling, zig-zag, or very fast?	3 s	Chooses the first test
2	**First differences** (term2 − term1, …)	10 s	Constant → AP (Type 1)
3	**Second differences** (difference of the differences)	10 s	Constant / squares / doubling → Type 2
4	Ratios: ×2, ×3 … then adjust by ± k (find k from the first pair)	10 s	Type 3
5	Compare each term with nearest square / cube	5 s	Type 5
6	Alternating jumps (big, small, big…) or zig-zag	5 s	Type 4 or Type 8 (interleaved)
7	Sum / product of the previous terms	5 s	Type 6

2.2 What you see → what to test first

What you see	Test first	Example
Slow rise, gaps getting bigger evenly	Second difference constant	9, 14, 24, 39, 59
Gaps themselves roughly double	Differences in GP, or ×2 − k	16, 24, 40, 72, 136
Each term about twice (or 3×) the previous	×r ± k	5, 15, 35, 75, 155
Big jump, small jump, big jump …	Alternating × and +	6, 18, 22, 110, 116
Up, down, up, down (zig-zag)	Two interleaved series	5, 3, 10, 7, 15, 11
Terms sit just beside perfect squares/cubes	$n^2 \pm k$, $n^3 \pm k$	3, 8, 15, 24, 35
Each term ≈ 1.6 × previous	Sum of previous two	5, 7, 12, 19, 31
Growth speeds up sharply, multiplier grows	×1, ×2, ×3 …	2, 2, 4, 12, 48, 240

2.3 Reference tables to memorise

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225
n^3	1	8	27	64	125	216	343	512	729	1000	1331	1728	2197	2744	3375
n	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
n^2	256	289	324	361	400	441	484	529	576	625	676	729	784	841	900

Set	Values
Primes up to 100 (25)	2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97
Powers of 2	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048
Powers of 3 / 4 / 5	3, 9, 27, 81, 243, 729 | 4, 16, 64, 256 | 5, 25, 125, 625
Factorials	1! = 1, 2! = 2, 3! = 6, 4! = 24, 5! = 120, 6! = 720
n(n+1)	2, 6, 12, 20, 30, 42, 56, 72, 90, 110, 132, 156

CHAPTER 3

Number Series: 9 Types

Rule, how to spot, shortcut and step-by-step PYQ solutions

3.1 Type 1: Constant difference (AP) [basic]

Rule: add or subtract the same number every time. **Spot:** first differences are all equal. Variant: two alternating differences (2, 5, 7, 10, 12 → +3, +2, +3, +2). **Example:** 7, 12, 17, 22, ? → difference 5 → **27**.

3.2 Type 2: Difference of differences [★★★ most tested]

Sub-type	How the differences behave	Next step	PYQ
2A AP of differences	5, 10, 15, 20 (rise by a constant)	add the same constant again	5
2B GP of differences	8, 16, 32, 64 (multiply by a constant)	multiply again	1
2C Special differences	squares 1, 4, 9, 16 / cubes 1, 8, 27 / primes	continue the family	0 (likely)
2D Third level	differences irregular, but their differences are squares	go one level deeper	1

Fast formula: **next term = last term + (last difference + second difference)**.

All five 2A questions at a glance

PYQ	Series	1st differences	Next diff	Answer
Eng 2019 Q37	9, 14, 24, 39, 59	5, 10, 15, 20	25	59 + 25 = **84**
Eng 2019 Q40	3, 17, 45, 87, 143	14, 28, 42, 56 (14×1, ×2, ×3, ×4)	70	143 + 70 = **213**
Maths 2024 Q78	11, 16, 26, 41, 61	5, 10, 15, 20	25	61 + 25 = **86**
Science 2024 Q83	112, 120, 130, 142, 156	8, 10, 12, 14	16	156 + 16 = **172**
SST 2023 Q20	24, 39, 64, 99, 144	15, 25, 35, 45	55	144 + 55 = **199**

PYQ Eng 2019 Q37 Type 2A, constant second difference

Q. 9, 14, 24, 39, 59, ?

Step 1. Write the differences under the series: 5, 10, 15, 20.
Step 2. These differences rise by 5 each time (second difference = 5), so the next difference is 20 + 5 = 25.
Step 3. Next term = 59 + 25 = 84.
Answer: 84

PYQ Eng 2019 Q40 Type 2A, differences are multiples of one number

Q. 3, 17, 45, 87, 143, ?

Step 1. Differences: 14, 28, 42, 56.
Step 2. Each is a multiple of 14 (14×1, 14×2, 14×3, 14×4), so the next difference is 14×5 = 70.
Step 3. 143 + 70 = 213.
Answer: 213

PYQ Maths 2024 Q91 Type 2B, differences in GP

Q. 16, 24, 40, 72, 136, ?

Step 1. Differences: 8, 16, 32, 64: each is double the previous, so the next is 128.
Step 2. 136 + 128 = 264.
Step 3. Cross-check as ×2 − 8: 16×2−8 = 24, 24×2−8 = 40, 40×2−8 = 72, 72×2−8 = 136, 136×2−8 = 264 ✓.
Answer: 264

PYQ SST 2023 Q3 Type 2D, third-level pattern

Q. 33, 44, 59, 83, 123, ?

Step 1. First differences: 11, 15, 24, 40 (irregular).
Step 2. Differences of these: 4, 9, 16 = 2^2, 3^2, 4^2. The next must be $5^2 = 25$.
Step 3. So the next first difference = 40 + 25 = 65, and 123 + 65 = 188.
Answer: 188

SHORTCUT: If the first differences look random, **always compute one more level** before giving up. Differences that are squares (4, 9, 16, 25) or cubes are the usual second-level families.

3.3 Type 3: Multiply and adjust (×r ± k) [★★ expected]

Form	Rule	Example	Next
3A GP	×r every time	3, 12, 48, 192	768
3B ×r ± k	multiply, then add/subtract a fixed number	5, 15, 35, 75, 155 (×2 + 5)	—
3C ×r ± changing number	multiply, then adjust by 1, 2, 3 … or 3, 4, 5 …	15, 27, 50, 95, 184 (×2 − 3, −4, −5, −6)	361
3D Rising multiplier	×1, ×2, ×3 … (factorial-like)	2, 2, 4, 12, 48, 240	1440

SHORTCUT: **Finding r and k in 10 seconds:** guess r from the growth (about double → 2). Compute k = (2nd term) − r × (1st term). Test the same k on the next pair. Example 5 → 15: 15 − 2×5 = 5, so k = +5; check 15 → 35: 15×2 + 5 = 35 ✓. If k changes by a steady step, it is type 3C.

3.4 Type 4: Alternating operations [★★]

Spot: a big jump then a small jump, repeating. **Rule:** two operations alternate (usually × then +), and the numbers used often rise by 1 each step.

PYQ Eng 2019 Q33 Alternating × and +

Q. 6, 18, 22, 110, 116, 812, ?

Step 1. 6 → 18 is ×3; 18 → 22 is +4; 22 → 110 is ×5; 110 → 116 is +6; 116 → 812 is ×7.
Step 2. The numbers used run 3, 4, 5, 6, 7 and the operations alternate × , + , × , + , ×. The next is +8.
Step 3. 812 + 8 = 820.
Answer: 820

TRAP: The last jump (812 → ?) is a **small** step. Options such as 890 and 905 are too big and can be dropped immediately.

3.5 Type 5: Squares, cubes and polynomial families [★ expected]

Family	Example	Family	Example
n^2	4, 9, 16, 25, 36	n^3	8, 27, 64, 125
$n^2 - 1$	3, 8, 15, 24, 35	$n^3 - 1$	0, 7, 26, 63, 124
$n^2 + 1$	2, 5, 10, 17, 26	$n^3 + 1$	2, 9, 28, 65, 126
$n^2 + n = n(n+1)$	2, 6, 12, 20, 30	squares of primes	4, 9, 25, 49, 121
$n^2 - n = n(n-1)$	0, 2, 6, 12, 20	$2^n \pm k$	3, 5, 9, 17, 33

SHORTCUT: For each term write the **nearest perfect square or cube**. If the gap is the same every time, that is the family (24 − 25 = −1, 35 − 36 = −1, so $n^2 - 1$).

3.6 Type 6: Sum of previous terms (Fibonacci-type) [★ expected]

Rule: each term = sum of the previous two (sometimes ± k, or sum of the previous three). **Spot:** from the third term on, $term_3 = term_1 + term_2$. **Example:** 5, 7, 12, 19, 31, 50, ? → 31 + 50 = **81**. Variants: 1, 3, 4, 7, 11

(Lucas); 1, 2, 4, 7, 13, 24 (sum of three).

3.7 Type 7: Prime-based [★ expected]

Patterns: the primes themselves; primes + k (3, 5, 9, 13 = prime + 1); squares of primes (4, 9, 25, 49); differences that are consecutive primes (2, 5, 10, 17, 28: +3, +5, +7, +11). **Spot:** irregular but small gaps and odd terms. Keep the prime list of 2.3 in mind.

3.8 Type 8: Two interleaved series [★ expected]

Spot: the series goes up, down, up, down, or no single rule works. **Method:** split into odd-position and even-position terms and solve each as its own series. **Example:** 5, 3, 10, 7, 15, 11, 20, ? → odd positions 5, 10, 15, 20 (+5); even positions 3, 7, 11, ? (+4) → **15**. The two halves may use different rule types (for example +5 and ×2).

3.9 Type 9: Wrong or odd term [★★ asked in SST 2023]

Step	What to do
1	Find the rule from the **first 3 or 4 terms** (or the middle terms) that look consistent.
2	Apply the rule forward; the first term that does not match is the wrong one.
3	Confirm: the term after it should fit the **correct** value, not the wrong one.

PYQ SST 2023 Q11 Identify the number that does NOT belong

Q. `15, 27, 50, 95, 184, 366`

Step 1. 15 → 27: 15×2 − 3 = 27. Next: 27×2 − 4 = 50. Then 50×2 − 5 = 95, and 95×2 − 6 = 184.
Step 2. The subtracted number rises 3, 4, 5, 6, so the next is 184×2 − 7 = 361.
Step 3. The series gives 366, not 361, so 366 is the wrong term.
Answer: 366

TRAP: The wrong term need not be the last one. In 2, 5, 11, 23, **48**, 95 the rule is ×2 + 1; 48 should be 47, and 95 = 47×2 + 1 continues from the **correct** value. Do not mark 95 as wrong.

CHAPTER 4

Letter & Alphanumeric Series

7 of the 16 PYQs: the column method solves almost all of them

4.1 Alphabet positions

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

Row 1: letters, row 2: forward position, row 3: backward position. **Opposite letters add up to 27** (A–Z, B–Y, C–X). Landmarks: **E=5, J=10, O=15, T=20, Y=25** (vowels A E I O U = 1, 5, 9, 15, 21). **Wrap-around:** after Z comes A, so X + 4 = B (24 + 4 = 28 → 28 − 26 = 2); before A comes Z (1 − 3 = −2 → −2 + 26 = 24 = X).

4.2 The column method (works for every cluster question)

Step	Action
1	Write every cluster as numbers (H = 8, S = 19 …) in rows under each other.
2	Read **each column as its own number series** (constant shift, rising shift, falling shift, alternating).
3	Extend each column one step; convert back to letters (add or subtract 26 if needed).

Typical column rules: constant ±k; shift rising (+1, +2, +3 …); shift falling; alternating ±; same letter repeated.

4.3 Find the next cluster

PYQ Eng 2022 Q41 Letter cluster, find next

Q. `HSD, KPI, NMN, QJS, ?`

Step 1. Numbers: HSD = 8, 19, 4; KPI = 11, 16, 9; NMN = 14, 13, 14; QJS = 17, 10, 19.
Step 2. Column 1: 8, 11, 14, 17 (+3). Column 2: 19, 16, 13, 10 (−3). Column 3: 4, 9, 14, 19 (+5).
Step 3. Next: 20, 7, 24 = T, G, X.
Answer: TGX

4.4 Alphanumeric clusters (letters and numbers together)

PYQ Eng 2022 Q56 Letter-number-letter, find next

Q. `H22N, I19J, K18G, N19E, ?`

Step 1. Column 1 (letters): H, I, K, N = 8, 9, 11, 14, shifts +1, +2, +3, so next shift +4 gives 18 = R.
Step 2. Column 2 (numbers): 22, 19, 18, 19, shifts −3, −1, +1, so next shift +3 gives 22.
Step 3. Column 3 (letters): N, J, G, E = 14, 10, 7, 5, shifts −4, −3, −2, so next shift −1 gives 4 = D.
Answer: R22D

PYQ SST 2023 Q15 Four letters + number

Q. `FWCU-5, IZZR-15, LCWO-35, ?, RIQI-155`

Step 1. Letters, position 1: F, I, L, [?], R (+3) so the missing one is O. Position 2: W, Z, C, [?], I (+3) so F.
Step 2. Position 3: C, Z, W, [?], Q (−3) so T. Position 4: U, R, O, [?], I (−3) so L. Cluster = OFTL.
Step 3. Number: 5, 15, 35, [?], 155 follows ×2 + 5 (35×2 + 5 = 75; 75×2 + 5 = 155 ✓).
Answer: OFTL-75 (Option 3)

SHORTCUT: When letters or numbers are given in the options, the first letter or two usually eliminates 2–3 options without finishing every column.

4.5 Incorrect / odd cluster (3 PYQs)

Run the column method on **all clusters**. The column that breaks at exactly one cluster identifies the odd one. Start with the column whose rule is easiest to see.

PYQ Maths 2024 Q80 Only one cluster is incorrect

Q. XQS VOQ TMO RKM PIL NGI

Step 1. Column 1: X, V, T, R, P, N: −2 each time ✓. Column 2: Q, O, M, K, I, G: −2 each time ✓.
Step 2. Column 3: S, Q, O, M, K, I should be the pattern (−2), so the 5th cluster should end in K, but PIL ends in L.
Step 3. The incorrect cluster is PIL (it should be PIK).
Answer: PIL (Option C)

PYQ Maths 2024 Q99 Only one cluster is incorrect

Q. SUX QSW OQT MOR KMP IKN

Step 1. Column 1: S, Q, O, M, K, I (−2) ✓. Column 2: U, S, Q, O, M, K (−2) ✓.
Step 2. Column 3 in numbers: 24, 23, 20, 18, 16, 14. From T (20) onward it falls by 2, so before T it should be 22 = V, not W.
Step 3. Cluster 2 should be QSV. QSW is incorrect.
Answer: QSW (Option A)

PYQ Science 2024 Q73 Select the letter-cluster that does NOT belong

Q. AWL XQG SKD OEZ KYV GSR

Step 1. Column 1: A, ?, S, O, K, G falls by 4 each time (S, O, K, G). Going back, 19 + 4 = 23 = W; and A (1) − 4 wraps to 23 = W. So the 2nd cluster must start with W, not X.
Step 2. Column 3: L, ?, D, Z, V, R falls by 4 (D, Z, V, R), so the 2nd must have H (not G).
Step 3. Column 2 (W, Q, K, E, Y, S: −6) is fine. XQG breaks two columns; the correct cluster would be WQH.
Answer: XQG (Option 2)

TRAP: Wrap-around is the favourite hidden step. In Science 2024 Q73, A → W needs 1 − 4 = −3, then +26 = 23. Always add 26 when you go below 1.

4.6 Repeating letter block with blanks

PYQ SST 2023 Q14 Fill the blanks sequentially

Q. P _ M R _ E _ F M _ W _ P F _ R _ E

Step 1. Seven blanks, and the same block repeats. The first full block has 6 letters: P _ M R _ E. Cut the series every 6 letters: (P _ M R _ E) (_ F M _ W _) (P F _ R _ E).
Step 2. Stack the blocks and read column-wise: position 1 = P, position 2 = F, position 3 = M, position 4 = R, position 5 = W, position 6 = E. The block is **PFMRWE**.
Step 3. Fill the blanks in order: F, W, P, R, E, M, W.
Answer: FWPREMW (Option 3)

SHORTCUT: Find a block length where a few letters repeat at the same positions; then stack the blocks like rows of a table. The blanks fill themselves from the other rows.

CHAPTER 5

Shortcuts and Traps

Marks saved and marks lost

5.1 Ten speed shortcuts

- **Write differences under the series** before thinking. It decides about 60% of questions in seconds.
- **Next term = last + (last diff + second diff)** for type 2A, with no need to extend the whole row.
- **Last-digit elimination:** work out only the last digit of the answer and cut options. For 59 → ?, adding 25 ends in 4: only 84 among 84, 80, 90, 79.
- **Growth check:** if the answer must be about double the last term and the option is 10% bigger, drop it.
- **Differences that are multiples of one number** (14, 28, 42, 56) are 2A: divide by the common factor.
- **Doubling differences** mean the series is 2^n + constant. Subtract the constant and the powers of 2 appear.
- **Square/cube gap test:** subtract the nearest square from each term; a constant remainder is the rule.
- **Letters: convert to numbers once**, solve columns, convert back only the answer.
- **Opposite letter = 27 − position** when a rule uses mirror letters.
- **In wrong-term questions** fix the rule from the first three terms, then test the rest.

5.2 Traps

Trap	How to avoid it
Stopping at the first difference	If differences are not equal, take differences again. 5 of 9 PYQs needed this.
Using a rule that fits only some terms	A valid rule must hold for **every** transition. Check at least the last two.
Reading the question wrong	‘Next term’, ‘wrong term’ and ‘does NOT belong’ need different answers for the same series.
Marking the last term wrong	The wrong term can be anywhere; the next term follows the **correct** value.
Forgetting to wrap letters	Below A add 26; above Z subtract 26.
Treating a zig-zag as one series	If no single rule works, split into odd and even positions.
Assuming one step for all columns in a cluster	Each column has its own rule (+3, −3, +5 in HSD KPI NMN).
Spending over a minute	After 40 seconds, eliminate options by last digit and move on.

CHAPTER 6

Practice Set

28 original questions in exam pattern; answers on the next page

Section A: Number series (find the next / wrong term)

1	7, 11, 19, 31, 47, ?	2	5, 6, 8, 11, 15, 20, ?
3	6, 11, 21, 41, 81, ?	4	2, 4, 10, 28, 82, ?
5	2, 7, 22, 67, 202, ?	6	2, 6, 12, 20, 30, 42, ?
7	3, 4, 8, 17, 33, 58, ?	8	1, 2, 10, 37, 101, ?
9	1, 3, 4, 7, 11, 18, ?	10	4, 12, 16, 48, 52, ?
11	2, 2, 4, 12, 48, 240, ?	12	3, 8, 15, 24, 35, 48, ?
13	5, 3, 10, 7, 15, 11, 20, ?	14	4, 9, 25, 49, 121, ?
15	6, 7, 9, 13, 21, ?	16	3, 6, 18, 72, 360, ?
17	1, 2, 4, 7, 13, 24, ?	18	20, 22, 26, 34, 50, ?
19	Wrong term: 3, 10, 29, 66, 129, 218	20	Wrong term: 2, 5, 11, 23, 48, 95

Section B: Letter and alphanumeric series

21	B, E, H, K, N, ?	22	AZ, BY, CX, DW, ?
23	ACE, FHJ, KMO, PRT, ?	24	ZA, YC, XE, WG, ?
25	Odd cluster: DXA, FVC, HTE, JRH, LPI	26	B2D, D6G, F18J, H54M, ?
27	Blanks: a _ c d a b _ d a b c _ a b c d	28	AZ, CX, FU, JQ, ?

Answer key with logic

Q	Answer	Logic
1	67	diffs 4, 8, 12, 16, then 20
2	26	diffs 1, 2, 3, 4, then 5
3	161	×2 − 1
4	244	×3 − 2
5	607	×3 + 1
6	56	n(n+1): 7 × 8
7	94	diffs 1^2, 2^2, 3^2, 4^2, 5^2, then $6^2 = 36$
8	226	diffs 1^3, 2^3, 3^3, 4^3, then $5^3 = 125$
9	29	sum of previous two: 11 + 18
10	156	×3, +4 alternately; 52 × 3
11	1440	×1, ×2, ×3, ×4, ×5, then ×6
12	63	$n^2 - 1$ with n = 8
13	15	odd positions +5 (5, 10, 15, 20); even +4 (3, 7, 11, 15)
14	169	squares of primes: 13^2
15	37	diffs 1, 2, 4, 8, then 16
16	2160	×2, ×3, ×4, ×5, then ×6
17	44	sum of previous three: 7 + 13 + 24
18	82	diffs 2, 4, 8, 16, then 32
19	129	$n^3 + 2$: $5^3 + 2 = 127$, so 129 is wrong
20	48	×2 + 1: 23×2 + 1 = 47, so 48 is wrong
21	Q	+3 each: N (14) + 3 = 17
22	EV	first +1 (A→E), second −1 (Z→V)
23	UWY	within block +2; block starts F, K, P, then U
24	VI	first −1 (Z, Y, X, W, V); second +2 (A, C, E, G, I)
25	JRH	col 1 +2, col 2 −2, col 3 +2: A, C, E, G, I, so JRH should be JRG
26	J162P	first +2; number ×3; last +3 (D, G, J, M, P)
27	b, c, d	block abcd repeats
28	OL	first +2, +3, +4, then +5 (J → O); second −2, −3, −4, then −5 (Q → L)

CHAPTER 7

Quick Revision

One page for the last day

If you see…	Try	Remember
Gaps growing evenly	2A: differences rise by constant	next = last + (last diff + d2)
Gaps doubling	2B: 2^n + c, or ×2 − k	16, 24, 40, 72, 136 → 264
Gaps irregular	2D: go one level deeper	2nd diffs 4, 9, 16 → 25
Roughly ×2 or ×3	3: ×r ± k, find k from first pair	5, 15, 35, 75, 155
Big jump, small jump	4: × then + with rising numbers	×3, +4, ×5, +6, ×7, +8
Near squares or cubes	5: constant gap from n^2, n^3	$n^2 - 1$: 3, 8, 15, 24
Each = sum of earlier two	6: Fibonacci-type	5, 7, 12, 19, 31
Primes or squares of primes	7: prime list	4, 9, 25, 49, 121
Zig-zag	8: split odd / even positions	5, 3, 10, 7, 15, 11
'Wrong / does not belong'	9: rule from first terms, test forward	×2 − 3, −4, −5, −6
Letter clusters	Columns as numbers	TGX, R22D, PIL, QSW, XQG
Letters with blanks	Find repeating block, stack rows	PFMRWE → F W P R E M W

PYQ answers to memorise (they repeat in new papers as patterns, not as exact questions)

Pattern	Differences / rule seen in PYQs
Differences 5, 10, 15, 20 …	Eng 2019 Q37 (9 → 84) and Maths 2024 Q78 (11 → 86)
Differences 8, 10, 12, 14 …	Science 2024 Q83 (112 → 172)
Differences 15, 25, 35, 45 …	SST 2023 Q20 (24 → 199)
Differences 14, 28, 42, 56 …	Eng 2019 Q40 (3 → 213)
Differences ×2 each	Maths 2024 Q91 (16 → 264)
Second differences 2^2, 3^2, 4^2	SST 2023 Q3 (33 → 188)
Letter shifts: +3, −3, +5	Eng 2022 Q41
Wrap-around falling shift −4	Science 2024 Q73

ROUTINE: Exam-day routine: differences → second differences → ratios (×r ± k) → squares/cubes → alternate/interleaved. For letters: numbers → columns → wrap-around. 40 seconds per question; skip and return if stuck.